

Nmap Fundamentals

A Quick, Non-Technical Reference Sheet

1 What is Nmap?

Nmap (Network Mapper) is a free tool used to discover devices on a network and find out what services/ports are open on them. It's one of the most widely used tools in networking and cybersecurity for scanning, auditing, and troubleshooting.

Think of it like knocking on every door of a building to see which ones are unlocked (open) and what's behind them.

Common uses:

- Discovering devices on a network
- Checking which ports/services are open on a machine
- Identifying operating systems and software versions
- Basic security auditing (finding weak points before attackers do)

2 Basic Syntax

```
nmap [scan type] [options] [target]
```

Example:

```
nmap 192.168.1.1
```

Scans the default set of common ports on that IP address.

3 Specifying Targets

Command	What It Does
<code>nmap 192.168.1.1</code>	Scan a single IP
<code>nmap 192.168.1.1-50</code>	Scan a range of IPs
<code>nmap 192.168.1.0/24</code>	Scan an entire subnet (all 254 devices)
<code>nmap example.com</code>	Scan by domain name
<code>nmap -iL targets.txt</code>	Scan a list of targets from a file

4 Common Scan Types

Flag	Scan Type
-sS	SYN scan (“stealth” scan) – fast, doesn’t complete full connection
-sT	TCP connect scan – completes full connection, more detectable
-sU	UDP scan – checks UDP ports (slower)
-sV	Version detection – identifies software/version running on open ports
-O	OS detection – guesses the target’s operating system
-sn	Ping scan – just checks which hosts are online, no port scan
-A	Aggressive scan – combines OS detection, version detection, script scanning

5 Port Options

Flag	What It Does
-p 80	Scan only port 80
-p 1-100	Scan a port range
-p-	Scan all 65,535 ports
-p 80,443,22	Scan specific listed ports
-F	Fast scan – only the 100 most common ports

6 Output Options

Flag	What It Does
-oN file.txt	Save results in normal readable format
-oX file.xml	Save results as XML
-oG file.txt	Save in “grepable” format (easy to search with grep)
-v	Verbose – show more detail while scanning

7 Timing & Speed

Nmap lets you control how fast/aggressive a scan is using -T0 (slowest, sneakiest) to -T5 (fastest, most detectable).

```
nmap -T4 192.168.1.1
```

-T4 is a common balanced choice – fast but not overly noisy.

8 Example Commands

Command	Purpose
nmap -sn 192.168.1.0/24	Find all live devices on a network
nmap -sV -p 22,80,443 192.168.1.1	Check specific ports and identify running services
nmap -A -T4 example.com	Full detailed scan (OS, versions, scripts)
nmap -p- 192.168.1.1	Scan every port on a target

9 Reading the Output

A typical result shows each port’s state:

- **open** – something is actively listening (e.g. a web server)
- **closed** – port reachable but nothing is listening
- **filtered** – a firewall is blocking Nmap from telling if it's open

10 Ethics & Legality

Only scan networks and devices you own or have explicit permission to test. Scanning networks without authorization can be illegal in many countries, even if no harm is intended.

11 Quick Recap

Nmap = tool to discover devices and open ports on a network

-sS/-sT = scan types (SYN vs full connect)

-sV = detect service versions

-O = detect operating system

-p = choose which ports to scan

-A = aggressive, all-in-one detailed scan

-T0 to -T5 = scan speed/stealth level

Always scan responsibly and with permission.